Day 5: Troubleshooting Network Services

DNS and DHCP Troubleshooting

Objective

The objective of this assignment is to identify and fix common DNS and DHCP problems in a local or simulated network.

Theory

DNS (Domain Name System) translates domain names such as example.com into IP addresses. DHCP (Dynamic Host Configuration Protocol) automatically provides a client with network settings such as an IP address, subnet mask, default gateway, and DNS server.

An IPv4 APIPA address usually belongs to 169.254.0.0/16. Windows may assign it when it cannot obtain an IPv4 address from a DHCP server. A device with an APIPA address normally cannot communicate properly outside its local link.

Task 1: Simulate a DNS Resolution Failure with the Hosts File

Safety note: This changes name resolution only on your own computer. Back up the hosts file first, use a clearly incorrect test entry, and remove it immediately after testing. Do not change shared or managed computers.

1. Open Notepad as Administrator.
2. Open the file: C:\Windows\System32\drivers\etc\hosts (select All Files if the file is not visible).
3. Make a backup copy of the hosts file. Add a temporary line using a documentation-only wrong address, for example: 192.0.2.123 www.spotify.com
4. Save the file. Open Command Prompt and run: ipconfig /flushdns
5. Run: ping www.spotify.com. Record the exact output. Depending on the Windows version and network, the ping may show the chosen address and time out, or another message; do not copy an expected result as if it were your actual output.
6. Remove the temporary line from the hosts file, save it, and run ipconfig /flushdns again. Verify that the temporary override is gone.

Important: A wrong hosts-file entry is a local name-resolution override, not a DNS-server failure. The ping may still resolve to the wrong address and then time out. Record exactly what your machine shows.

Task 2: Incorrect Default Gateway in a Simulated DHCP Network

7. Open Cisco Packet Tracer or another network simulator. Add a router or Layer 3 device, a switch, a server configured for DHCP, and one client PC.
8. Configure the server's DHCP pool with a valid network, subnet mask, address range, DNS server, and an intentionally incorrect default gateway in the pool's gateway field.
9. Set the client to obtain its IP configuration automatically (DHCP). Renew the lease or toggle DHCP if needed.
10. On a Packet Tracer PC, open Desktop > IP Configuration and record the assigned IP address, subnet mask, and default gateway. On a Windows client, use ipconfig /all.
11. Test communication to another device in the same subnet, then test communication to a destination outside the subnet if your simulation includes one. A wrong gateway may still allow same-subnet communication but prevent off-subnet access.
12. Fix the gateway value in the DHCP pool to the router interface address for that client subnet. Renew the client lease and verify the corrected gateway.

Task 3: Websites Fail by Name but Work by IP

Scenario: A client can reach a known server by IP address, but a website name does not resolve.

13. Check basic connectivity to the correct destination IP address. Use a known test destination in your lab.
14. Run nslookup example.com (replace example.com with the lab domain or a domain you are permitted to test). Record the DNS server shown and the response or error.
15. Run nslookup example.com <DNS-server-IP> to query the intended DNS server directly, where supported.
16. On Linux, use dig example.com and dig @<DNS-server-IP> example.com. On Windows, use ipconfig /all to inspect the configured DNS servers.
17. Check whether the configured DNS server is reachable, whether its service is running, and whether the record/forwarder is correct. Check firewall rules for DNS traffic (usually UDP/TCP port 53).
18. Correct the DNS server address, DNS record, or forwarder as appropriate. Flush the local cache with ipconfig /flushdns on Windows, then repeat nslookup and the website test.

Task 4: Client Receives an APIPA Address (169.254.x.x)

Possible causes

- DHCP server is powered off, stopped, or unreachable.
- Client and server are in different VLANs/subnets and DHCP relay (IP helper) is missing or incorrect.
- DHCP scope has no free addresses or its pool is configured incorrectly.
- Wrong VLAN, disconnected cable, Wi-Fi association problem, or switch port issue.
- Firewall or ACL blocks DHCP messages; DHCP generally uses UDP ports 67 (server) and 68 (client).
- Incorrect DHCP pool network, subnet mask, exclusions, or address range.

Step-by-step resolution

19. Check the cable/Wi-Fi connection, link status, switch port, and client VLAN.
20. Run ipconfig /all on Windows and confirm the address is 169.254.x.x and DHCP is enabled.
21. Run ipconfig /release followed by ipconfig /renew. Record any error message.
22. Check that the DHCP server/service is running and reachable. In the simulator, inspect the DHCP service and pool settings.
23. Confirm the pool has free addresses and that the network, subnet mask, gateway, and exclusions are correct.
24. If the client is on another subnet, verify that the router interface and DHCP relay/IP helper are configured correctly.
25. Review firewall/ACL rules and ensure DHCP traffic is not blocked.
26. Renew the client lease again and verify it receives an address from the intended subnet, plus the correct gateway and DNS server.

Task 5: Common DNS and DHCP Issues Table

Protocol / Issue	Symptoms	Possible causes	Recommended solution
DNS server unreachable	Names fail to resolve; nslookup times out	Wrong DNS IP, server down, routing/firewall issue	Check ipconfig /all, server status and reachability; correct DNS IP or restore service.
Incorrect or missing DNS record	One domain resolves incorrectly or returns NXDOMAIN	Wrong A/CNAME record or missing zone record	Check authoritative zone/record, correct it, then test with nslookup/dig.
Stale local DNS cache or hosts override	A name resolves to an old or unexpected IP	Cached answer or hosts-file entry	Inspect hosts file, remove incorrect override, run ipconfig /flushdns and query again.
DHCP server unavailable	Client gets 169.254.x.x or no lease	Service/server down or network path blocked	Check link/VLAN, DHCP service, firewall and relay; renew the lease.
DHCP scope exhausted	New clients cannot obtain a lease	All usable addresses are leased or pool too small	Inspect active leases, release unused leases or expand the

			scope safely.
Incorrect gateway or scope options	Client gets an IP but cannot reach other networks or uses wrong DNS	Wrong gateway/DNS option or wrong subnet mask	Correct DHCP pool options, renew the lease, and verify with ipconfig /all.

Conclusion

DNS problems affect name resolution, while DHCP problems affect automatic IP configuration. Troubleshooting should begin with the client configuration and network connectivity, then check the relevant server, scope/records, routing, and firewall rules. After applying a fix, repeat the test and record the actual result.

Practical Evidence Checklist

☐ Hosts-file test output recorded and temporary line removed

☐ Incorrect and corrected DHCP gateway documented in the simulator

☐ nslookup or dig test output recorded

☐ APIPA causes and resolution steps documented

☐ Screenshots added from my own tests or simulator

Submission note: Fill in actual outputs and screenshots after doing the practical. The sample scenario and command guidance are not proof that a test was performed.